

Advice for Applying Machine Learning

Evaluating Models, Choosing Among Them, and Diagnosing Bias and Variance

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Week 6 · Part 1

From building models to improving them

Weeks 1–5 gave us

Linear and logistic regression, regularisation, and neural networks trained with Adam. You can now build a model for almost any supervised task.

Building is the easy half. When the first model disappoints — and it usually does — there are a dozen things you could try, and some of them cost months. This week is about choosing *which* one, from evidence rather than hunches.

Today

1. Deciding what to try next; evaluating a model
2. Model selection: training, validation and test sets
3. Diagnosing bias and variance; the role of λ
4. Baselines and learning curves
5. What to try next, revisited; neural networks

Part 2 covers the development process itself: error analysis, adding data, transfer learning, deployment, fairness, and metrics for skewed classes.

Evaluating a Model

Your model makes large errors. What next?

Things you could try

1. Get more training examples.
2. Try a smaller set of features.
3. Get additional features.
4. Add polynomial features (x_1^2, x_2^2, $x_1 x_2$, ...).
5. Decrease the regularisation parameter λ.
6. Increase λ.

Each is reasonable. Each helps some models and is useless for others. Collecting more data alone can take a team months.

Diagnostic

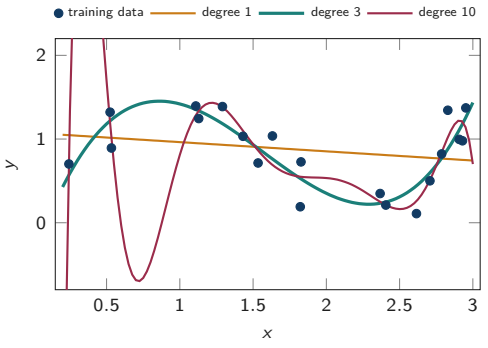
A test you run to learn what is and is not working in a learning algorithm, so that you can decide how to improve it.

Key idea

Diagnostics take time to run, but much less time than pursuing the wrong fix. The rest of this lecture builds the two most useful ones: the **bias/variance** diagnosis and the **learning curve**.

The running example: a synthetic regression task

20 training points; $y = \sin 2x + 0.5x$ plus noise, $\sigma = 0.25$



Setting

50 points from a known curve plus Gaussian noise: 20 for training, 15 for validation, 15 for testing. Models: polynomials of degree $d = 1, \ldots, 10$. Cost $J = \frac{1}{2m} \sum_i (f(x^{(i)}) - y^{(i)})^2$.

What the picture shows

Degree 1 misses the curve. Degree 10 passes close to every training point and swings wildly between them — off the chart near $x = 0.3$. Degree 3 looks right.

Plots do not scale

With one feature we can look. With 64 pixels or 40 weather stations we cannot. We need a number that says "degree 10 is bad".

Evaluating on data the model has not seen

Split the data

Train on one part (often 70–80%); hold back the rest as a **test set**, used only to measure.

$$J_{\text{test}} = \frac{1}{2m_{\text{test}}} \sum_{i=1}^{m_{\text{test}}} \left(f(\mathbf{x}_{\text{test}}^{(i)}) - y_{\text{test}}^{(i)} \right)^2$$

Compute it *without* the regularisation term.

For classification

J_{test} can be the logistic cost, but the more readable measure is the **misclassification error**: the fraction of test examples with $\hat{y} \neq y$.

Our example, degree 10

J_{train}	0.0164
J_{test}	0.3362

A training cost this low says the curve threads the training points. The test cost, twenty times larger, says it has memorised them.

Key idea

J_{train} measures fit. Only error on held-out data estimates how the model will do on new examples — its **generalisation error**.

Model Selection and Cross-Validation

Choosing the degree: the wrong way

Tempting procedure

Fit $d = 1, 2, \ldots, 10$ on the training set. Compute J_{test} for each. Pick the d with the smallest J_{test} and report that J_{test}.

The test set has been used to choose

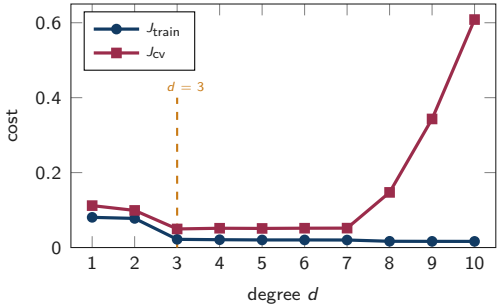
d is now a parameter fitted to the test set, exactly as $\mathbf{w}$ was fitted to the training set. The reported J_{test} is the *best of ten tries* and is optimistic. In our data it would pick $d = 7$ (0.0380), the luckiest of several near-equal degrees.

Three-way split

- **Training set** (e.g. 60%): fit $\mathbf{w}, b$.
- **Cross-validation set** (validation, dev set; 20%): choose d, λ, the architecture.
- **Test set** (20%): report once, at the end.

Each set has one job. Any decision taken by looking at a set makes that set's error an underestimate.

Choosing the degree: the right way



Procedure

1. Fit each degree on the training set.
2. Compute J_{cv} for each; pick the smallest: $d = 3$, $J_{\text{cv}} = 0.0496$.
3. Report J_{test} for $d = 3$ only: **0.0429**.

Two curves, two stories

J_{train} only falls as d grows: more flexibility always fits the training data better. J_{cv} falls, flattens, then rises — the classic U. The bottom of the U is the model to choose.

Same procedure for anything you choose: the number of layers and units in a network, λ, the learning rate.

Diagnosing Bias and Variance

Bias and variance, read from two numbers

Diagnosis	J_{train}	J_{cv}	Our example
High bias (underfit)	high	high, $\approx J_{\text{train}}$	$d = 1$: 0.0806 and 0.1119
High variance (overfit)	low	$\gg J_{\text{train}}$	$d = 10$: 0.0164 and 0.6085
Just right	low	low, near J_{train}	$d = 3$: 0.0219 and 0.0496
Both	high	$\gg J_{\text{train}}$	rarer with one feature; common in networks

Bias

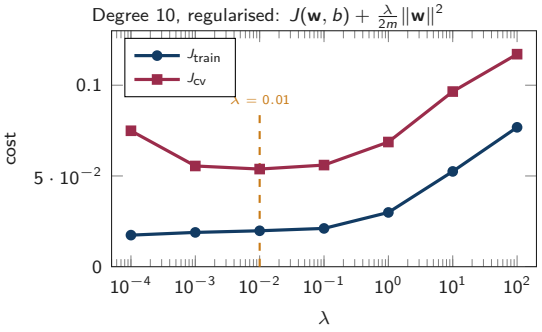
The model is too simple to represent the pattern, so it does poorly even on the data it was trained on. More of the same data will not help.

Variance

The model is so flexible that it fits the noise in its particular training set. A different training set would give a very different model.

"High" and "low" need a reference point: the baseline, later in this lecture.

Regularisation moves a model along the same axis



Reading the curve

Small λ (left): the degree-10 polynomial is nearly free — high variance. At $\lambda = 0$, $J_{\text{cv}} = 0.6085$, off the chart.

Large λ (right): weights are forced towards 0, f flattens — high bias.

In between, J_{cv} is smallest at $\lambda = 0.01$ (0.0538); its test cost is 0.0397.

Mirror image

Against d, variance grows to the right. Against λ, variance grows to the *left*. Choose λ exactly as d: try a range roughly doubling or tenfold, and take the smallest J_{cv}.

Baselines and Learning Curves

"High" compared with what? Establish a baseline

Speech recognition (illustrative)

Some audio is so noisy that people cannot transcribe it either.

	case A	case B	case C
human error	10.6%	10.6%	10.6%
training error	10.8%	15.0%	15.0%
validation error	14.8%	15.5%	19.7%
bias gap	0.2	4.4	4.4
variance gap	4.0	0.5	4.7
diagnosis	variance	bias	both

Baseline level of performance

The error you can reasonably hope to reach: human-level performance, a competing system, or past experience.

Two gaps

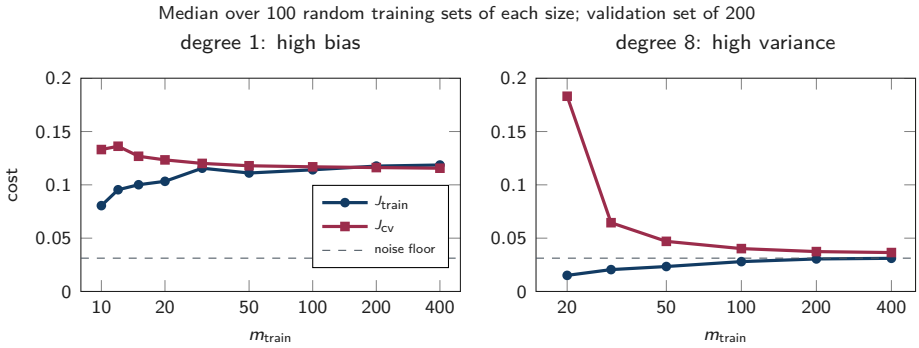
bias gap = training error − baseline

variance gap = validation error − training error

Our example has a perfect baseline

The noise has $\sigma = 0.25$, so no model can beat $J = \sigma^2/2 = 0.03125$ on average. Degree 3's training cost, 0.0219, is *below* it: the fit has absorbed some noise.

Learning curves: error against training-set size



High bias. Both curves flatten at about 0.117, far above the floor, and meet. From $m = 100$ to $m = 400$, J_{cv} improves by 0.001: more data does not help.

High variance. A large gap at small m that closes as m grows; J_{cv} falls from 0.183 to 0.037, approaching the floor: more data helps.

What to Try Next, and Neural Networks

Deciding what to try next, revisited

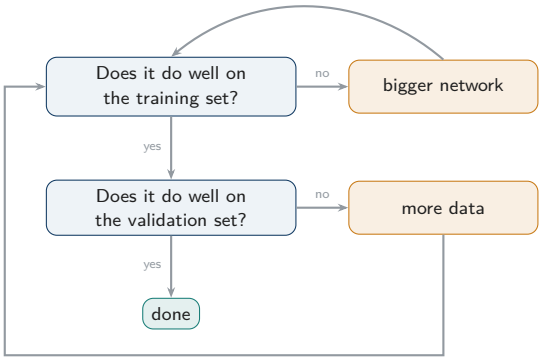
Action	Fixes
Get more training examples	high variance
Try a smaller set of features	high variance
Increase λ	high variance
Get additional features	high bias
Add polynomial features	high bias
Decrease λ	high bias

Diagnose first, then act

Compute J_{train}, J_{cv} and a baseline. Whichever gap is larger tells you which half of the table to use.

Do not shrink the training set

Fewer examples make J_{train} look better (the left end of a learning curve) but never fix bias — and they make variance worse.



A recipe that rarely existed before

Large networks trained on modest data are usually *low-bias*. So: grow the network until training error is acceptable, then add data until validation error is. Bias and variance can be attacked one at a time.

Key idea

A larger, well-regularised network almost never does worse than a smaller one. Its costs are computation and data, not accuracy.

```
Dense(units=25, activation='relu',
      kernel_regularizer=
          tf.keras.regularizers.L2(0.01))
```

Wrap-up and Assessment

Summary

What to remember

- Judge a model on data it was not trained on; J_{train} only measures fit.
- Train / cross-validation / test: fit, choose, report. Never choose on the test set.
- High bias: J_{train} high. High variance: $J_{\text{cv}} \gg J_{\text{train}}$. Measure both against a baseline.
- J_{cv} is U-shaped in d and in λ (mirrored).
- Learning curves: bias plateaus, more data does not help; variance gap closes, more data helps.
- Neural networks: bigger network for bias, more data for variance.

Formulas

$$J_{\text{cv}} = \frac{1}{2m_{\text{cv}}} \sum_i \left(f(\mathbf{x}_{\text{cv}}^{(i)}) - y_{\text{cv}}^{(i)} \right)^2$$

$$\text{bias gap} = J_{\text{train}} - \text{baseline}$$

$$\text{variance gap} = J_{\text{cv}} - J_{\text{train}}$$

Next

Part 2: the iterative loop of development, error analysis, adding data, transfer learning, deployment, fairness, and precision and recall.

Descriptive questions

1. List six actions you might take when a model makes large errors. Explain why a diagnostic should come first. **[5]**
2. Why is J_{train} a poor estimate of a model's performance on new data? Define J_{test} for regression and the misclassification error for classification. **[5]**
3. Explain why selecting the polynomial degree with the test set makes the reported test error optimistic. State the role of each part of a three-way split. **[6]**
4. Sketch J_{train} and J_{cv} against polynomial degree, label the high-bias and high-variance regions, and explain the shape of each curve. **[6]**
5. Using the slide's numbers, diagnose degrees 1, 3 and 10 of the running example. **[5]**
6. Sketch J_{train} and J_{cv} against λ and explain why it is a mirror image of the degree plot. **[6]**

Descriptive questions (continued)

7. Define a baseline level of performance and give three ways of obtaining one. **[4]**
8. Human error is 2.0%, training error 2.3%, validation error 7.1%. Compute both gaps, give the diagnosis, and name two remedies. **[6]**
9. Draw typical learning curves for a high-bias and a high-variance model. For each, state whether collecting more data will help, and why. **[8]**
10. For each action in the "what to try next" table, state which problem it addresses and explain why. **[6]**
11. Describe the bias/variance recipe for neural networks and explain why it did not apply to small, classical models. **[5]**
12. In the running example, degree 3 reaches $J_{\text{train}} = 0.0219$, below the noise floor of 0.03125. Explain how this is possible and what it implies. **[6]**

Design questions

1. **Design an evaluation protocol.** A Chattogram hospital has 3000 labelled chest X-rays from 900 patients and wants to choose among five network architectures and three values of λ. Design the split (including how you keep one patient's images together), the selection procedure, and what you report. **[12]**
2. **Design a diagnosis.** A crop-yield model has training $R^2 = 0.62$ and validation $R^2 = 0.60$, while an agronomist's estimate achieves 0.75. Diagnose the model, and design the next three experiments in priority order, with the result that would make you stop each one. **[10]**
3. **Design a data-collection decision.** Collecting 1000 more labelled examples will cost Tk 5 lakh. Design an experiment, using only the data you already have, that predicts whether it is worth paying for. **[12]**
4. **Design a λ search.** Design a procedure to choose λ for a network whose training takes four hours, under a one-week compute budget. Justify your grid and how you will use partial results. **[10]**

Reading

Primary text

G. James, D. Witten, T. Hastie and R. Tibshirani, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021.

→ **§2.2.2** — the bias–variance trade-off; **§5.1** — the validation-set approach and k-fold cross-validation; **§6.2.1** — choosing the ridge λ.

Supplementary

C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.

→ **§1.3** — model selection and cross-validation; **§3.2** — the bias–variance decomposition.

Lab (CSE 816, Week 6): model evaluation and selection; diagnosing bias and variance.